

Infinitely many primes ending in a 7

Theorem 1. *There are infinitely many primes that end in a 7. i.e. there are infinitely many primes that are congruent to 7 mod 10*

To prove this, we will need to consider special series, now known as Dirichlet L-series, and a special linear combination of their logarithms. In our special case, these simplify greatly and tests for convergence become much easier. It is helpful to begin with choosing primitive roots to simplify our notation.

7 is a primitive root mod 10, namely $(\mathbb{Z}/10)^\times = \{1, 3, 7, 9\} = \{7^0, 7^3, 7^1, 7^2\}$

Also, we will use the 4th roots of unity (since $4 = \phi(10)$). To do this succinctly, we can note that i is a primitive 4th root of unity, namely $\{1, i, -1, -i\} = \{i^0, i^1, i^2, i^3\}$.

Now, we work towards the definition of our specific Dirichlet L-series. Consider that for $s > 1$, by geometric series

$$\frac{1}{1 - i^{k\gamma_p} \frac{1}{p^s}} = 1 + i^{k\gamma_p} \frac{1}{p^s} + i^{2k\gamma_p} \frac{1}{p^{2s}} + i^{3k\gamma_p} \frac{1}{p^{3s}} + \dots$$

where $p \neq 2, 5$ and γ_p is defined to be such that $p \equiv 7^{\gamma_p} \pmod{10}$.

If we take the product of this over all primes $p \neq 2, 5$ we get

$$\prod_{p \neq 2, 5} \frac{1}{1 - i^{k\gamma_p} \frac{1}{p^s}} = \sum_n i^{k\gamma_n} \frac{1}{n^s} \tag{1}$$

where the sum is taken over $n \equiv 1, 3, 7, 9 \pmod{10}$ and γ_n is such that $n \equiv 7^{\gamma_n} \pmod{10}$.

To see this, note that for $n = q_1^{m_1} q_2^{m_2} q_3^{m_3} \dots$, where $q_i \neq 2, 5$ are distinct primes and $m_i > 0$, the n th term of the product is of the form $i^{k(m_1\gamma_{q_1} + m_2\gamma_{q_2} + m_3\gamma_{q_3} + \dots)} \frac{1}{n^s}$. By the definition of γ_n , $m_1\gamma_{q_1} + m_2\gamma_{q_2} + m_3\gamma_{q_3} + \dots \equiv \gamma_n \pmod{4}$, and thus by Euler's theorem, this general exponent simplifies as required.

Define $L_k = \sum_n i^{k\gamma_n} \frac{1}{n^s}$ where the sum is taken over $n \equiv 1, 3, 7, 9 \pmod{10}$ for $k \in \{0, 1, 2, 3\}$. To develop a proof, we must study the behavior of these series as $s \rightarrow 1^+$.

Lemma 1. $L_0 \rightarrow \infty$ as $s \rightarrow 1^+$

Proof. By (1) we have that

$$L_0 = \prod_{p \neq 2,5} \frac{1}{1 - \frac{1}{p^s}} = (1 - \frac{1}{2^s})(1 - \frac{1}{5^s}) \prod_p \frac{1}{1 - \frac{1}{p^s}} = (1 - \frac{1}{2^s})(1 - \frac{1}{5^s}) \sum_n \frac{1}{n^s}$$

Where n is now taken over all positive integers.

As $s \rightarrow 1^+$, the RHS approaches the harmonic series, multiplied by a constant, hence it diverges to ∞ as required. $\square$

Lemma 2. L_2 converges to a positive number as $s \rightarrow 1^+$.

Proof. Consider directly when $s = 1$. We have, calculating γ_n in each case,

$$\begin{aligned} L_2 &= \sum_{n \equiv 1 \pmod{10}} \frac{1}{n} - \sum_{n \equiv 3 \pmod{10}} \frac{1}{n} - \sum_{n \equiv 7 \pmod{10}} \frac{1}{n} + \sum_{n \equiv 9 \pmod{10}} \frac{1}{n} \\ &= (1 - \frac{1}{3} - \frac{1}{7} + \frac{1}{9}) + (\frac{1}{11} - \frac{1}{13} - \frac{1}{17} + \frac{1}{19}) + \dots \\ &= 1 - (\frac{1}{3} + \frac{1}{7}) + (\frac{1}{9} + \frac{1}{11}) - (\frac{1}{13} + \frac{1}{17}) + \dots \\ &= 1 - \frac{10}{21} + \frac{20}{99} - \frac{30}{221} + \dots \end{aligned}$$

Which converges by the alternating series test, hence L_0 converges when $s = 1$. Furthermore, the partial sums are always strictly positive, so the limit when $s = 1$ of L_0 is also strictly positive. $\square$

Lemma 3. L_1 and L_3 converges to non-zero numbers as $s \rightarrow 1^+$.

Proof. Again, consider directly when $s = 1$. We have, calculating γ_n in each case,

$$\begin{aligned} L_1 &= \sum_{n \equiv 1 \pmod{10}} \frac{1}{n} + i \sum_{n \equiv 3 \pmod{10}} \frac{1}{n} - i \sum_{n \equiv 7 \pmod{10}} \frac{1}{n} - \sum_{n \equiv 9 \pmod{10}} \frac{1}{n} \\ &= \sum_{n \equiv 1 \pmod{10}} \frac{1}{n} - \sum_{n \equiv 9 \pmod{10}} \frac{1}{n} + i(\sum_{n \equiv 3 \pmod{10}} \frac{1}{n} - \sum_{n \equiv 7 \pmod{10}} \frac{1}{n}) \\ L_3 &= \sum_{n \equiv 1 \pmod{10}} \frac{1}{n} - i \sum_{n \equiv 3 \pmod{10}} \frac{1}{n} + i \sum_{n \equiv 7 \pmod{10}} \frac{1}{n} - \sum_{n \equiv 9 \pmod{10}} \frac{1}{n} \\ &= \sum_{n \equiv 1 \pmod{10}} \frac{1}{n} - \sum_{n \equiv 9 \pmod{10}} \frac{1}{n} - i(\sum_{n \equiv 3 \pmod{10}} \frac{1}{n} - \sum_{n \equiv 7 \pmod{10}} \frac{1}{n}) \end{aligned}$$

Therefore, $L_3 = \overline{L_1}$ and it suffices to show that L_1 converges. To do this, we can show that its real and imaginary parts both converge using the alternating series test.

$$\begin{aligned} \operatorname{Re}(L_1) &= \sum_{n \equiv 1 \pmod{10}} \frac{1}{n} - \sum_{n \equiv 9 \pmod{10}} \frac{1}{n} \\ &= 1 - \frac{1}{9} + \frac{1}{11} - \frac{1}{9} + \frac{1}{21} - \frac{1}{29} + \dots \\ \operatorname{Im} L_1 &= \sum_{n \equiv 3 \pmod{10}} \frac{1}{n} - \sum_{n \equiv 7 \pmod{10}} \frac{1}{n} \\ &= \frac{1}{3} - \frac{1}{7} + \frac{1}{13} - \frac{1}{17} + \frac{1}{23} - \frac{1}{27} + \dots \end{aligned}$$

These both converge by alternating series test, hence both L_1 and L_3 converge when $s = 1$.

To show that the limits of L_1 and L_3 are non zero, consider the product

$$L_1 L_3 = L_1 \overline{L_1} = \operatorname{Re}(L_1)^2 + \operatorname{Im}(L_1)^2 \geq 0$$

Now, the partial sums of $\operatorname{Re}(L_1)$ are strictly positive, hence $\operatorname{Re}(L_1) > 0$ and thus $L_1 L_3 > 0$. Hence L_1 and L_3 converge to a non-zero number as required. $\square$

Before we prove the theorem, we still need one more critical tool; the logarithm. Let us find the general formula of $\log L_k$ for $k \in \{0, 1, 2, 3\}$ using Taylor's theorem. For $s > 1$

$$\log L_k = \log \prod_{p \neq 2, 5} \frac{1}{1 - i^{k\gamma_p} \frac{1}{p^s}} = \sum_{p \neq 2, 5} \log \frac{1}{1 - i^{k\gamma_p} \frac{1}{p^s}}$$

Note that by Taylor's theorem

$$-\log(1 - x) = x + \frac{1}{2}x^2 + \frac{1}{3}x^3 + \dots$$

Hence the above can be rewritten as

$$\begin{aligned} \log L_k &= \sum_{p \neq 2, 5} \left(i^{k\gamma_p} \frac{1}{p^s} + \frac{1}{2} i^{2k\gamma_p} \frac{1}{p^{2s}} + \frac{1}{3} i^{3k\gamma_p} \frac{1}{p^{3s}} + \dots \right) \\ &= \sum_{p \neq 2, 5} i^{k\gamma_p} \frac{1}{p^s} + \frac{1}{2} \sum_{p \neq 2, 5} i^{2k\gamma_p} \frac{1}{p^{2s}} + \frac{1}{3} \sum_{p \neq 2, 5} i^{3k\gamma_p} \frac{1}{p^{3s}} + \dots \end{aligned} \tag{2}$$

Proof. (Infinitely many primes ending in 7)

Consider the linear combination

$$\begin{aligned} \log L_0 + i^{-1} \log L_1 + i^{-2} \log L_2 + i^{-3} \log L_3 &= \sum_{p \neq 2,5} (1 + i^{\gamma_p-1} + i^{2(\gamma_p-1)} + i^{3(\gamma_p-1)}) \frac{1}{p^s} \\ &\quad + \frac{1}{2} \sum_{p \neq 2,5} (1 + i^{2\gamma_p-1} + i^{2(2\gamma_p-1)} + i^{3(2\gamma_p-1)}) \frac{1}{p^{2s}} \\ &\quad + \frac{1}{3} \sum_{p \neq 2,5} (1 + i^{3\gamma_p-1} + i^{2(3\gamma_p-1)} + i^{3(3\gamma_p-1)}) \frac{1}{p^{3s}} \\ &\quad + \dots \end{aligned}$$

Note that

$$1 + i^{h\gamma_p-1} + i^{2(h\gamma_p-1)} + i^{3(h\gamma_p-1)} = 0$$

except when $h\gamma_p-1 \equiv 0 \pmod{4}$ where it is 4. This is equivalent to when $p^h \equiv 7 \pmod{10}$.

Therefore, the linear combination simplifies to

$$\begin{aligned} &\log L_0 + i^{-1} \log L_1 + i^{-2} \log L_2 + i^{-3} \log L_3 \\ &= 4 \left(\sum_{p \equiv 7 \pmod{10}} \frac{1}{p^s} + \frac{1}{2} \sum_{p^2 \equiv 7 \pmod{10}} \frac{1}{p^{2s}} + \frac{1}{3} \sum_{p^3 \equiv 7 \pmod{10}} \frac{1}{p^{3s}} + \dots \right) \end{aligned} \quad (3)$$

As $s \rightarrow 1^+$, $\log L_0 \rightarrow \infty$ by Lemma 1 and $i^{-1} \log L_1 + i^{-2} \log L_2 + i^{-3} \log L_3$ is finite by Lemmas 2 and 3. Hence the LHS of (3) tends to ∞ and thus so too must the RHS.

It is known that the series $\frac{1}{2} \sum_{p^2 \equiv 7 \pmod{10}} \frac{1}{p^{2s}} + \frac{1}{3} \sum_{p^3 \equiv 7 \pmod{10}} \frac{1}{p^{3s}} + \dots$ converge to finite values as $s \rightarrow 1^+$. In fact, this is true for the series over all positive integers n , which clearly bounds our case from above. Thus, it must be that the term $\sum_{p \equiv 7 \pmod{10}} \frac{1}{p^s} \rightarrow \infty$ as $s \rightarrow 1^+$.

This concludes our proof since for this series to diverge, there must be infinite terms. So, we have that there are infinitely many primes congruent to $7 \pmod{10}$ and hence primes ending in a 7. $\square$

This result can be generalised to all arithmetic sequences of the form $a + kd$ where $k \in \mathbb{N}$ and a, d are coprime. To do this, it is best to begin with the case of d being prime. In this case, both a primitive root mod p and a primitive $p - 1$ th root of unity exist and our method works excluding the tests for convergence, since the series likely do not simplify into alternating series in any way. To test convergence, further methods in complex analysis such as use of the gamma function are required.

Once the proof has been made for prime differences, it extends quite naturally to prime powers for odd primes due to the existence of a primitive root in that case. It can then be extended to all integers by slightly altering the indexes used, since a primitive root is no longer guaranteed. This is slightly more tedious and one must account for powers of 2 slightly differently as well.